

Order Flow and Expected Option Returns

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ABSTRACT

I show that the inventory risk faced by market-makers has a first-order effect on option prices. I introduce a simple approach that decomposes the price impact of trades into inventory risk and asymmetric information components. While both components are large for option trades, the inventory risk component is larger. Using the full panel of daily option returns, I find that option order imbalances attributable to inventory risk have five times larger impact on option prices than previously thought. Finally, I find that past order imbalances have greater predictive power than any other commonly used predictor of option returns.

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In this paper, I study how the inventory risk that market-makers face affects prices in the equity options market. These variables have a well-established theoretical relation. Market-makers are required to provide liquidity and accommodate customer order imbalances and thus their position often deviates substantially from the desired level. In turn, a risk-averse investor requires a higher expected return on his inventory as compensation for holding a portfolio with suboptimal weights. The extent of inventory risk is proportional to the position's size and volatility, the market-maker's risk aversion, and the expected holding period. Option market-makers face inventory risk because, in practice, options cannot be fully replicated by trading in the underlying.

I show that option market-maker inventory risk has a first-order effect on option prices. This conclusion is important for at least two reasons. First, the equity options market is economically large: options on about 1.5 billion shares are traded daily in the U.S. Second, option prices are extensively used to measure variables ranging from equity return volatility to the stochastic discount factor, and these measures may be biased if the inventory risk component is not removed from option prices. Thus, a new generation of option pricing models that accounts for inventory risk is needed.

The importance of inventory risk for the options market suggests that its role in other markets may be more important than previously thought.¹ Prior literature shows that buying pressure is associated with higher option prices, but its economic magnitude is small compared to other factors, making inventory risk appear to be of secondary importance. The difference in the results is due to two methodological issues that this paper solves. First, order flow and prices are endogenous, so concurrent factors such as news affect both. The popular approach of regressing option returns on same-day order

imbalances can thus produce biased coefficients. Second, besides having an inventory risk impact, order imbalance also reflects informed trading (Shleifer (1986)) and correlates with changes in economic fundamentals. This problem is commonly ignored by attributing the entire order imbalance to only one of these three factors.² I find that after addressing these problems, the market-wide order imbalance has a particularly large effect on option prices, as market-makers manage inventory risk on a portfolio basis.

The paper's main conclusion is supported by two sets of results. First, I establish the importance of inventory risk at the intraday level. Using a novel methodology, I find that the inventory risk component of the price impact of option trades is larger than the asymmetric information component for each stock in my sample. Second, using a full panel of daily option returns, I establish that inventory risk is a dominant factor at the daily level. The instrumental variables (IV) approach identifies the inventory-risk component of price pressure by studying how past order option imbalances predict future imbalances. According to the IV approach, a typical inventory shock moves option prices by 3.7% the same day, which is five times larger than the change implied by conventional OLS. Finally, I conduct a direct horse-race between more than 50 commonly used predictors of future option returns including the inventory-related order imbalance, I find that order imbalance has the highest predictive power by a large margin. Economically, if it increases by one standard deviation, the next-day return is 1% higher.

With respect to the intraday analysis, the interaction between trades and quotes is key to understanding how and why prices change. The literature identifies two reasons why quoted prices increase after a buyer-initiated trade. First, market-makers adjust upward their beliefs about fair value as the trade may contain private information (e.g.,

Glosten and Milgrom (1985)). Second, market-makers require compensation for allowing their inventory position to deviate from the desired level, and thus a risk-averse market-maker will accommodate a subsequent buy order only at a higher price (e.g., Stoll (1978)). Both arguments imply that quotes change in the direction of trade, but for different reasons.

Building on Huang and Stoll (1997), I introduce a novel microstructure method for evaluating the size and relative importance of information asymmetry and dynamic inventory control for price dynamics. The method decomposes the price impact of trades into an inventory risk component and an asymmetric information component. The main idea is that investors instantly receive identical information about a trade, while only the market-maker at the exchange where the trade is executed experiences a change in inventory. Thus, the price response at the trading exchange includes both the asymmetric information and the inventory risk components, while the price impacts at the non-trading exchanges contain only the asymmetric information component. The difference between the price responses of the trading and nontrading exchanges identifies the inventory risk component. Both price impact components can be easily estimated empirically from observed price responses. I formally implement this idea by extending the framework of Madhavan and Smidt (1991) to multiple competitive market-makers.

This new method is needed because, as Huang and Stoll (1997) discuss, existing methods struggle to separate the information and inventory components.³ These methods commonly assume that permanent price changes are attributed to asymmetric information, while the impact of inventory risk is temporary and is reversed in a matter of minutes. However, empirical evidence suggests that although the effect of inventory risk

on prices is by definition temporary, sometimes it takes weeks to unwind. Thus, at the intraday level, the effects of inventory risk and asymmetric information are largely permanent, making it hard for conventional methods to separate them. The method that I introduce avoids this criticism as it makes no assumption about how long it takes for the inventory effect to disappear. Overall, this method contributes to the literature on the role and measurement of market-maker inventory risk and to the literature on distinguishing the inventory and asymmetric information components of bid-ask spreads.

I apply this new method to the equity options market using tick-level data for options on 39 stocks over the period April 2003 to October 2006, when the options market had already converted to its modern form with predominantly electronic trading. The method yields several results. First, contrary to Vijh (1990), I find that option trades have a significant price impact. Thus, option trades contain a lot of new information that drives changes in option prices. Second, although both price impacts have a significant effect, inventory risk has larger price impact (0.4%) than asymmetric information (0.2%). Moreover, the inventory risk component is larger for each stock in my sample. Thus, at least within my sample, inventory risk plays a dominant role in option price formation. This result implies that while option market-makers are concerned about trading against informed investors, they are even more concerned about inventory control. Third, both price impacts are increasing and concave in trade size. Thus, large trades are more informative than medium and small trades, suggesting that option investors no longer engage in “stealth trading,” whereby informed investors split their option trades and medium-sized trades are the most informed (Anand and Chakravarty (2007)). Fourth, the underlying stock price responds to option trades instantly, and the price impact of option

trades on the underlying stock price is permanent and increasing in trade size. Option trades are thus informative about the underlying stock price level. This result complements a large literature on informed trading in the options market that is based mostly on daily returns (e.g., Pan and Poteshman (2006)). Finally, both price impact components depend on option and trade parameters. In particular, both price impacts are larger for out-of-the-money and short-term options, and buyer-initiated trades have a larger impact particularly for the inventory risk component. Most of these relations are intuitive. For example, out-of-the-money options provide the highest leverage, which attracts informed investors. Overall, all these findings contribute to our understanding of the role of inventory risk and asymmetric information in the formation of option prices.

This paper employs one of the largest data sets in recent intraday options literature. However, several limitations of the sample must be acknowledged. First, almost all the sample stocks have very actively traded options. Second, the sample period corresponds to mostly a bull market with relatively little market stress. These observations are important because several papers on the components of the bid-ask spread in equities find substantial variation in their magnitudes depending on method, sample, and period. Also, the method that I introduce estimates the impact of inventory risk but is silent about the time horizon over which it disappears. However, the daily results help address this concern by showing that the effect of inventory risk effect on prices takes at least several days to unwind and is large for the entire universe of optionable stocks during a different sample period.

In the second set of tests, I extend the analysis to the daily level and again show that inventory risk is the main determinant of expected option returns. Using more than

five years of market-makers' daily order imbalance data from the International Securities Exchange, I study how inventory risk, measured as market-makers' order imbalance, affects returns on delta-neutral option portfolios (whose value is neutral to small changes in the underlying price). This analysis is based therefore on changes in inventory and prices rather than levels.

In the first of these tests I employ instrumental variables and find that order imbalances are persistent in the option market, with the expected future imbalances attributable to inventory risk rather than informed trading. I then quantify the effect of inventory-related imbalances on option prices. In particular, in the first stage of the 2SLS regression, order imbalance is separately instrumented by three nested sets of instruments based on option expiration dummy variables and past individual and market wide order imbalances. All of the coefficients in the first stage are highly significant, implying that order imbalances are persistent. That is, buying is followed by more buying on the next day. In the second stage, I regress option returns on the predicted imbalances and find similar coefficients across all instrument sets, confirming their validity. The estimates suggest that the effect of inventory risk on prices is large. If the inventory-related order imbalance increases by one standard deviation, option returns will be 3.7% higher on the same day while the OLS estimates will be five times smaller. The effect of inventory risk was previously underestimated because the role of market-wide order imbalance and the endogeneity problem were not considered.

Finally, in the second set of tests at the daily level, I conduct a horse-race between the inventory-related order imbalance and more than 50 commonly used predictors of future option returns. To my knowledge, this is the first such comprehensive comparison.

The inventory-related order imbalance is a clear winner in this contest, with an increase of one standard deviation leading to a 1% increase in next-day option returns. In addition, this return predictability (1) is observed multiple days into the future, (2) is robust to other ways of computing option returns, (3) is equally strong for stocks with liquid and illiquid options, and (4) is higher during the financial crisis, as inventory risk is more expensive to manage during periods of high volatility and risk aversion.

The daily analyses rely on expected order imbalances as a conceptually clean way to separate inventory risk from informed trading and changes in fundamentals. Both my empirical hypotheses and the order imbalance measure come directly from the Chordia and Subrahmanyam (2004) model applied to the options market.

Although option market-makers hedge dynamically with the underlying stock in the efforts to reduce inventory risk, this hedging only partially reduces the volatility of an option portfolio. In practice, options cannot be perfectly replicated and the residual volatility is large because volatility is stochastic, prices and volatility often jump, the stochastic process for the underlying is unknown, and perhaps most importantly, hedging is not possible when markets are closed overnight. A combination of factors makes inventory risk so significant for options: market-makers' combined capital is relatively small compared to the size of the market and the complexity of option risks, customer order flow is usually one-sided, which forces market-makers to hold inventory positions for a long time, and institutional restrictions limit the entry of new liquidity providers.

The remainder of the paper is organized as follows. In Section I I provide a brief review of the related options literature. In Section II I introduce the market microstructure method, while in Sections III and IV I apply it to the equity options market. Section V

establishes the importance of inventory risk at the daily level, and Section VI concludes. In Internet Appendix I conduct a number of robustness tests and evaluate the method's assumptions.⁴

I. Related Options Microstructure Literature

The literature on the interaction between quoted option prices and trades is scarce. The approach I use is closest in spirit and objective to that of Vijn (1990) and Berkman (1996), who study the price impact of large option trades. Vijn finds no price impact for a sample of 137 large option trades from CBOE in 1985, while Berkman finds similar results for 456 trades from EOE in Amsterdam in 1989. Since then, the option market has converted to electronic trading and has become many times more liquid. With a much larger sample, I demonstrate that the price impact is positive and significant. A similarity between our papers is that we study price impact directly for a “clean” subsample of trades. However, even after all the filters, my final sample contains more than a quarter of all trades, whereas Vijn and Berkman use a very small share of total option volume. In another related paper, Chan, Chung, and Fong (2002) conclude that all information in the options market is contained in quote revisions and none in option trades. In contrast, my results suggest that option trades are quite informative.

The paper is also related to the literature on the determinants of the bid-ask spread in the options market. I briefly review this literature here. Early papers document how the option bid-ask spread depends on option parameters. George and Longstaff (1993) show that the dollar bid-ask spread is increasing in option moneyness and time to expiration. In addition to this first-order effect, Jameson and Wilhelm (1992) find that the spread increases in measures of convexity such as option gamma and vega, but the convexity

explains a relatively small portion of the spread. These stylized facts are important but hard to interpret. The bid-ask spread consists of three components: fixed costs (also called “order processing costs”), inventory risk, and asymmetric information. Do in-the-money options have larger spreads because informed investors prefer them or because of higher inventory risk? The literature does not offer a conclusive answer, but the variation in fixed costs across option classes is the most likely explanation. The fixed cost of establishing the initial delta-hedge (Cho and Engle (1999)) and maintaining it (rebalancing costs in Boyle and Vorst (1992), Engle and Neri (2010)) can explain a significant portion of the spread for in-the-money and at-the-money options, respectively. For example, Kaul, Nimalendran, and Zhang (2004) find that initial hedging costs explain about half of the option spread. However, the costs of inventory risk remain largely unstudied. These costs should be distinguished from the replication (hedging) costs. For an extreme example, a risk-neutral market-maker is not concerned about inventory risk but incurs all of the replication fixed costs. Replication costs depend on the bid-ask spread of the underlying and the expected size of hedging trades, and thus are part of fixed cost, and affect the option bid-ask spread but not the quote midpoint. In contrast, inventory risk is driven by the residual volatility of the hedged option portfolio, and a market-maker’s inventory changes from trade to trade, inventory risk affects both the spread and the price.

Surprisingly few papers use structural approaches to estimate the components of option bid-ask spreads. Rather, the literature overwhelmingly follows a reduced-form approach: proxies such as option parameters, stock bid-ask spread (and its components), volatility, probability of informed trading (PIN), and their interactions are assigned to

components of the option bid-ask spread (e.g., PIN to the asymmetric information component), and their relative importance is inferred from regression analysis. This approach assumes that all the explanatory variables are exogenous and are perfect proxies for the spread components. My paper complements this literature by developing and applying a structural approach to options to infer the inventory risk and information components.

This paper also uses a much larger and recent data set than most of the option bid-ask spread literature. The literature typically relies on about one month of data from the late 1990s, when the options market had a very different market structure and faced lower liquidity. For example, Cho and Engle (1999) use 180,239 option trades from May 1993. Kaul, Nimalendran, and Zhang (2004) use 182,605 observations from February 1995. Engle and Neri (2010) use data on nine stocks for four days in 2007. In comparison, my final sample consists of more than 7.5 million option trades for 39 stocks over more than three years ending in October 2006. Engle and Neri rightly emphasize that the problem today is that option data “poses a technical problem in terms of computer power.”

II. Price Impact Decomposition Method

A. The Idea

This subsection explains the intuition behind the proposed method, while the next subsection introduces it formally. Consider three competitive market-makers, A, B, and C, each at a separate exchange, who quote bid and ask prices for the same security in a modern electronic market environment. Imagine that a buy order of size V^* arrives at exchange A, the “trading exchange,” and all three exchanges have been quoting exactly

the same ask price right before the trade. Figure 1 describes this setup graphically. Also, assume that the quoted size at the receiving exchange V^A is bigger than the trade size V^* – otherwise, the buy trade will mechanically increase the price at trading exchange A.

[Figure 1 here]

This trade changes the inventory position of market-maker A, who is the counterparty in this transaction, while the inventories of B and C are unchanged. Thus, only A's price response Δp^A contains the inventory risk component. However, because information in modern electronic markets is standardized, all three market-makers instantly learn the same information about the transaction. Thus, all three market-makers adjust their prices Δp^i to reflect the new information inferred from the trade, and all price responses contain the asymmetric information component. This intuition can be summarized as follows:

$$\begin{aligned}\Delta p^A &= PI(Inventory) + PI(Information) \\ \Delta p^B &= \Delta p^C = PI(Information) ,\end{aligned}\tag{1}$$

where $PI(Inventory)$ and $PI(Information)$ denote the price impacts of inventory risk and asymmetric information, and $\Delta p = p_{t+\Delta t}^i - p_t^i$ is the change in price for the i^{th} market maker between the pre-trade time t and time $t + \Delta t$. The evaluation period Δt is set to five seconds in most empirical tests, which gives market-makers more than enough time to respond but is small enough to limit interference between multiple trades. In a general specification, equation (1) will also contain an error term as price responses are affected by other factors such as public nontrade information and microstructure noise.

Equations (1) can be easily solved for the two price impact components:

$$PI(Inventory) = \Delta p^A - (\Delta p^B + \Delta p^C)/2$$

$$PI(Information) = (\Delta p^B + \Delta p^C)/2 . \quad (2)$$

The asymmetric information component is simply an average price response by nontrading market-makers, while the inventory risk component is the difference between the price responses of the trading and nontrading market-makers. This example considers a single trade, but because individual price responses are very noisy, taking an average over a large number of trades is required to estimate the components. Equation (1) will be generalized in the next subsection in equations (9) and (10) to account for trade direction, endogeneity between trades and quoted, and public news flow.

Overall, both price impact components are estimated by applying a modified version of equation (2) to the subsample of all trades for which multiple exchanges quote exactly the same price in the trade direction.

B. Conceptual Framework

In this subsection I introduce a theoretical framework that extends the model of Madhavan and Smidt (1991) to multiple competitive market-makers. I then show how the two price impact components are estimated within this framework in a way that is in line with the idea in equations (1) and (2). The exposition closely follows Madhavan, Richardson, and Roomans (1997).

Consider a market for a risky security whose fundamental value evolves over time. A large number of fully competitive market-makers continuously quote bid and ask prices at which investors can buy or sell X shares.⁵ Let μ_t denote market-makers' common belief about the fair value at time t . Changes in beliefs can arise from three sources: new public information unrelated to trading, information from trades, and slow diffusion of old public information.

First, the flow of public information to the market causes revisions in beliefs without any trading. Denote by ϵ_t the innovation in beliefs between time t and $t + \Delta t$ due to new public information. I assume that ϵ_t has zero mean and is uncorrelated with the trading process. Second, trades also provide information, as investors may trade on private information. Following client's purchase of an asset, market-makers increase their estimate of the asset's fair value by $\theta(x_t - E(x_t|F_t))$, where the unexpected order flow is multiplied by the information-asymmetry parameter $\theta \geq 0$. A linear price impact is commonly assumed in the literature. For example, Bagnoli, Viswanathan, and Holden (2001) find necessary and sufficient conditions for linearity, while numerous other papers show that it arises in particular models of interaction between uninformed and informed investors with market-makers. Finally, beliefs often respond slowly to public information. The expected change in beliefs $\Delta\mu_t$ predicted by stale public information F_t is denoted $E(\Delta\mu_t|F_t)$. Most models assume that all past public information is instantly reflected in prices and thus cannot predict future returns. Empirically, however, intraday returns are quite predictable. Including this feature makes the framework more realistic but the main conclusions do not depend on it. Combining the three components, market-makers' beliefs about fair value evolve according to

$$\mu_{t+\Delta t} - \mu_t = \Delta\mu_t = E(\Delta\mu_t|F_t) + \theta(x_t - E(x_t|F_t)) + \epsilon_t. \quad (3)$$

Let $p_{t,i}^a$ denote the pre-trade ask price set by the i^{th} market-maker at time t . Each market-maker sets bid and ask prices conditional on getting a trade. Market-makers share a common belief about the asset's fair value but have different desired and actual inventory positions $I_{d,i}$ and $I_{t,i}$. Following previous literature, price depends linearly on the current inventory's deviation from the desired level.⁶ Importantly, if the number of

market-makers is large and they are fully competitive, then there is little interaction between them. Market-makers set prices that make them indifferent between getting and not getting the next trade, and actions of a single market-maker have little effect on the state of the market. One of the limitations of my framework is that without this assumption the model becomes richer but analytically intractable.

Market-makers also face significant fixed costs of $\varphi \geq 0$ per share. Fixed costs are usually a major component of the bid-ask spread (Huang and Stoll (1997)), but they have little effect on quote midpoint dynamics. Microstructure noise caused by tick size and other frictions is captured by random variable $\vartheta_{t,i}$, which has zero mean and is uncorrelated with other covariates. Bid and ask prices are therefore summarized as follows:

$$p_{t,i}^b = \mu_t + \theta(-X - E(x_t|F_t)) - \gamma(I_{t,i} + X - I_{d,i}) - \varphi \cdot X + \vartheta_{t,i}^b \quad (4)$$

$$p_{t,i}^a = \mu_t + \theta(X - E(x_t|F_t)) - \gamma(I_{t,i} - X - I_{d,i}) + \varphi \cdot X + \vartheta_{t,i}^a,$$

where the bid-ask half-spread is equal to

$$ba_{t,i} = \frac{1}{2}(p_{t,i}^a - p_{t,i}^b) = (\theta + \gamma + \varphi) \cdot X + \frac{1}{2}(\vartheta_{t,i}^a - \vartheta_{t,i}^b). \quad (5)$$

Equation (5) is a classic decomposition of the bid-ask spread into asymmetric information θ , inventory risk γ , and fixed cost φ components. Several approaches are able to distinguish fixed costs from the rest of the bid-ask spread, but have difficulty distinguishing the inventory and information components. Huang and Stoll (1997) summarize this by saying that “estimates of adverse information probably include inventory effects as well since existing procedures cannot distinguish the two.” Interestingly, equation (4) implies that if the difference between actual and desired

inventory levels is large, inventory risk will cause prices to deviate substantially from fair value while having little effect on the bid-ask spread.

After following the standard steps in equations (3) to (5), I explicitly account for the difference in market-maker responses to trades. If t is the time of a trade and Δt is a small evaluation period that gives market-makers just enough time to respond to the trade, market-makers will set prices according to equations (4) and (5) at both t and $t + \Delta t$. However, the extent to which the ask price increases after a buyer-initiated trade differs between trading and nontrading market-makers. To develop the intuition, I first conduct the analysis only for a buy trade and ask price before generalizing the analysis at the end of this subsection. In the Internet Appendix I consider the case in which trade direction is misclassified for some trades. The trading market-maker gets extra inventory (X) and increases the ask price by

$$\Delta p_{t,i}^a = E(\Delta \mu_t | F_t) + \theta(X - E(x_t | F_t)) + \gamma \cdot X + \epsilon_{t+\Delta} + (\vartheta_{t+\Delta t,i}^a - \vartheta_{t,i}^a), \quad (6)$$

while nontrading market-makers get the same information but have no change in inventory, and thus they also increase their ask price but by less than the trading market-maker.⁷

$$\Delta p_{t,i-}^a = E(\Delta \mu_t | F_t) + \theta(X - E(x_t | F_t)) + \gamma \cdot 0 + \epsilon_{t+\Delta} + (\vartheta_{t+\Delta t,i-}^a - \vartheta_{t,i-}^a).$$

To simplify the notation, assume that buy and sell trades are conditionally equally likely, and thus $E(x_t | F_t) = 0$, and that the noise/error terms are combined into a single term $\varepsilon_{t+\Delta,i} = \epsilon_{t+\Delta} + (\vartheta_{t+\Delta t,i}^a - \vartheta_{t,i}^a)$ that has zero mean by construction. The price responses of the trading (with index “ i ”) and nontrading (“ $i-$ ”) market-makers in equation (6) can be then re-labeled with this simpler notation as follows:

$$\Delta p_{t,i}^a = E(\Delta \mu_t | F_t) + \theta X + \gamma X + \varepsilon_{t+\Delta,i} \quad (7)$$

$$\Delta p_{t,i-}^a = E(\Delta\mu_t|F_t) + \theta X + \varepsilon_{t+\Delta,i-}.$$

Then the difference between the price responses for trading and nontrading market-makers is

$$\Delta p_{t,i}^a - \Delta p_{t,i-}^a = \gamma X + (\varepsilon_{t+\Delta,i} - \varepsilon_{t+\Delta,i-}). \quad (8)$$

Price responses (7) derived within this general framework match the intuitive results of equation (1). Indeed, the price response of the trading exchange (7) contains both the inventory risk and the asymmetric information impacts, while the response of nontrading exchanges in equation (7) contains only the information impact θ . However, both price responses also depend on the expected changes in price $E(\Delta\mu_t|F_t)$ and microstructure noise ε_t . Microstructure noise ε has zero mean and thus can be eliminated by taking an average over a large number of trades. Note that the Law of Large Numbers requires that the average instead of median should be taken; the Internet Appendix explains that the distribution of ε is not symmetric due to price discreteness.

Unfortunately, the expected changes in price $E(\Delta\mu|F_t)$ cannot be averaged out in a similar way and should be estimated separately. Investors do not simply execute trades at random – they time their trades (Hasbrouck (1991)). In particular, they buy when the price is expected to increase anyway, and thus $E(\Delta\mu|F_t) > 0$ for buy trades and $E(\Delta\mu|F_t) < 0$ for sell trades. Muravyev and Pearson (2014) further explain why accounting for expected price changes is crucial.

The inventory risk and asymmetric information components of price impact can be estimated from equations (7) and (8) in a sufficiently large sample that the microstructure noise terms average to zero. I can now generalize the formulas to account for both buy and sell trades, which requires new notation. The trade direction indicator

I_i^{BS} is set to 1 for buy trades and -1 for sells. The price response $\Delta p_{t,i}^{BS}$ is computed based on ask prices for buys ($\Delta p_{t,i}^a$) and bid prices for sell trades ($\Delta p_{t,i}^b$). With this notation, the main equations for the price impact decomposition become

$$\text{Asymmetric information:} \quad \theta \cdot X = E[I_i^{BS} \cdot (\Delta p_{t,i}^{BS} - E(\Delta \mu_t | F_t))] \quad (9)$$

$$\text{Inventory risk:} \quad \gamma \cdot X = E[I_i^{BS} \cdot (\Delta p_{t,i}^{BS} - \Delta p_{t,i-}^{BS})] \quad (10)$$

The inventory risk component of price impact is simply the difference between the price responses of trading and nontrading market-makers adjusted for trade direction. The asymmetric information component is an average of the price responses of nontrading market-makers adjusted for the expected change in price due to slow diffusion of public information. Again, the intuition of equation (2) is preserved in this general framework.

In the following subsection I outline how the main equations (9) and (10) can be applied to the data. In particular, Section IV.C applies the general framework above to the options market. First, I choose particular methods for inferring trade direction I^{BS} and expected price changes $E(\Delta \mu_t | F_t)$. Standard algorithms (such as the quote rule) correctly classify the sign of the vast majority of trades. I estimate a regression model for expected quote changes on historical data and then apply the estimated model to the public information available right before a trade to produce the expected price change for this trade. I further explain this step in the Internet Appendix. Second, I select a subset of all trades that satisfies the method's assumptions ($\{i\} \subset I$). Specifically, I select trades with at least two exchanges quoting the same best price in the direction of the trade. I use these exchanges to compute price responses for a given trade. Third, for each trade i in this subset ($i \in \{i\}$), I compute the unexpected price change for nontrading exchanges $I_i^{BS} \cdot$

$(\Delta p_{t,i-}^{BS} - E(\Delta \mu_t | F_t))$ and the difference between the price responses of the trading and nontrading exchanges $I_i^{BS} \cdot (\Delta p_{t,i}^{BS} - \Delta p_{t,i-}^{BS})$, both of which are adjusted for trade sign I_i^{BS} . Finally, equations (9) and (10) imply that the average over all trades $\{i\}$ for these individual trade responses produces the final estimates of the information and inventory price impacts.

C. Assumptions

In this subsection I summarize the method's assumptions and checks that the options market satisfies them. The key assumptions are as follows:

- 1) Markets are transparent, information is spread instantly, and information is standardized, that is, everybody receives the same message about a trade.
- 2) Liquidity is provided primarily by market-makers, that is, most liquidity providers are concerned about inventory risk.
- 3) Market-makers do not actively share inventory directly with each other.
- 4) Many competitive market-makers operate in the market.
- 5) Multiple market-makers often quote the same best price.

Although a number of markets satisfy these assumptions, the equity options market fits them particularly well. First, in early 2003, prior to the start of my sample period, all options exchanges were connected through Linkage and the National Best Bid and Offer (NBBO) rule was introduced. At the same time, investors gained access to real-time information about the best prices from all exchanges. In the age of floor-based trading, market-makers on the exchange floor had better information about trades.⁸ However, this advantage diminished greatly in the modern electronic markets with anonymous trade counterparties. Everybody gets a standardized message with transaction

information in less than a second. All option market-making is electronic, and the prices are set by computer algorithms. The options market is at least as transparent and technologically developed as the equity market.

Second, market-makers stand on the liquidity-providing side of most trades. In the options market, market-makers transfer liquidity not only over time but also across different options. With several hundred option contracts available for each underlying, two investors rarely select the same option and thus they are likely to trade with a market maker. Also, exchange rules grant lead market-makers substantial competitive edge over other liquidity providers (e.g., the 60/40 NBBO order split rule). These rules further strengthen lead market-makers' position as the main liquidity providers and make it hard for new players to enter.

Third, market-makers do not share inventory in the options market. This practice, which was common in some over-the-counter markets where dealers could trade directly with each other, involves a large customer trade being followed by a sequence of smaller trades between dealers to share the inventory. In this case, all dealers get some inventory, and their price responses to a trade contain both information and inventory components. My method would underestimate the inventory risk component and overestimate the asymmetric information component in this case. The Internet Appendix shows that "hot-potato" trading is not common in the options market. Also, although a market-making firm often makes markets in multiple stocks and at multiple exchanges, it makes a market at only one exchange for options on any given stock.

Fourth, the method I introduce requires that many fully competitive market-makers operate in the market, otherwise the multiperiod model becomes intractable. The

strategic interaction is limited if market-makers are fully competitive, because in this case they post prices that make them indifferent between trading and not trading at any moment. The prices are set conditional on getting a trade in equation (4) and thus do not depend on the probability of getting it, that is, the quoted prices are the same irrespective of how many exchanges are at the top of the book.⁹ For instance, if three market-makers quote the same best ask price and a buy order arrives, two nontrading market-makers will increase the price not only because of asymmetric information but also because two market makers have more market power than three. This method would therefore underestimate the inventory risk component and overestimate the asymmetric information price impact in the case of imperfect competition. Empirically, competition is high in the options market – for most trades, at least four exchanges quote the best price. Also, instead of manually setting prices, market-makers rely on computer programs, making it harder to implement strategic interactions.

Firth, options exchanges quote the same price most of the time because of the large tick size. For example, SEC (2007) shows that at least three exchanges quote the same NBBO price more than 75% of the time. This number is consistent with sample summary statistics in Table I.

Finally, in the Internet Appendix I discuss empirical and theoretical implications of inferring trade direction on the results. In short, because the method focuses on trades with multiple exchanges quoting the trade price, my test design ensures that the number of misclassified trades is very small. Also, I show that trade misclassification does not affect the relative magnitude but will make both components smaller, and thus make it harder to find significant price impacts.

III. Data Description and Sample Construction

A. Data Description

The tick-level data were collected by Nanex, a firm specializing in selling real-time data feeds to proprietary traders. The data set contains all quotes and trades for 39 stocks including four exchange-traded funds (ETFs) and their options from April 2003 to October 2006. The selected stocks had the most liquid options based on option trading volume in March 2003, which includes large-capitalization technology stocks that have persistently high option volume, as well as a few smaller-capitalization stocks that happened to be of trading interest during the spring of 2003. The data were archived and time-stamped by Nanex to 25 millisecond precision as transactions arrived and come from all U.S. exchanges on which a given contract is traded. Nanex records the option part of the data from OPRA data stream. The transaction price, size, exchange code, and other information are available for each trade. OPRA does not report option trade direction, and thus I infer this by applying the quote rule to the NBBO. If the trade is at the midpoint of the NBBO, the quote rule is applied to the best bid offer (BBO) from the exchange at which the trade occurs. In the internet appendix, I argue that this algorithm has small estimation error. The exchange-level best quotes and volumes are available for quoted prices, that is, the data include each instance in which any exchange adjusts its best quote or quoted volume, even if this change does not alter NBBO. The data set also contains stock market data similar to TAQ. A more detailed description of this data set is provided by Muravyev, Pearson, and Broussard (2013).¹⁰

For the daily-level analysis, I compute order imbalances using the data obtained from the International Securities Exchange (ISE). This data set contains daily nonmarket-

maker volume for all options listed on ISE starting in May 2005. For each option, the daily trading volume and number of trades have four types: open buy (close buy), where investors buy options to open new (close old) position and open sell (close sell). Trading volume is also divided by investor type, which allows me to compute the order imbalance faced by market-makers. The data only include transactions that were executed at ISE; however, throughout most of the sample period, ISE was the largest equity options exchange with a market share of about 30%. OptionMetrics is a common source of price information on equity options. For each option contract, it contains end-of-day best bid and ask prices as well as other information such as volume, open interest, implied volatility, and option Greeks. Returns and volume for the underlying stocks are also taken from OptionMetrics to avoid data loss from merging with CRSP. After all filters, discussed next, the final sample contains more than 1000 stocks each trading day.

B. Data Filters

To construct a final sample that satisfies the method's requirements, I proceed in several steps. First, I employ a standard prescreening to eliminate illiquid and bizarre options. Trade transactions should satisfy the following conditions:

- (a) An option should have time to expiration between 10 and 400 calendar days.
- (b) Option moneyness measured by absolute option delta (i.e., the sensitivity of the option price to changes in the underlying price) is between 0.2 and 0.8, that is, options with at least some "optionality" are selected.
- (c) The transaction time is between 9:35 and 15:55. The first and last five minutes of trading are excluded to avoid open and close rotations.

In total, 15,298,816 out of 28,127,906 trade transactions satisfy these conditions. Dropping any of the above filters does not change the main results

I next apply the minimum conditions required by the method:

- (d) At least two exchanges (including the one receiving a trade) quote the trade price; otherwise, equation (10) is not well defined.
- (e) Trade size is smaller than the quoted size at the trading exchange. This condition avoids a mechanical price impact at the trading exchange.
- (f) Option bid price is larger than 10 cents to avoid outliers, as the price impact is normalized by the option price.

Condition (d) also serves another purpose. Since the trade price is equal to the NBBO price quoted by at least two exchanges, this condition resolves ambiguity about trade direction as further discussed in the Internet Appendix. Condition (e) makes it less likely for the trading exchange to move its quotes and, if anything, works against finding a significant inventory risk impact. Conditions (d) to (f) reduce the sample size to 7,684,944 trades. Finally, a few outliers with an absolute price impacts of more than 50% are removed, which results in a final sample of 7,684,040 trades. Thus, a significant portion of all option trades passes all the filters. Panel B of Table I summarizes the sample size at each screening stage.

C. Computation of Price Impact Components

The information risk and asymmetric inventory price impacts are estimated by applying equations (9) and (10) to the final sample of option trades discussed in the previous subsection. I follow the general outline described at the end of Section III.B. Specifically, I first process each trade in the final sample independently and then

calculate the average price responses over all trades. For each trade, I identify nontrading exchanges that together with the trading exchange quote the NBBO price in the trade direction (price responses of the other exchanges are not used to avoid endogeneity). These price responses are used to compute the components of the price impact in four steps. First, following the method described in the Internet Appendix, I compute the expected price changes, that is, by how much the relevant quoted price (bid for sell trades) would be expected to change if no trade arrived. Second, following equation (9), I compute the information price impact by subtracting the expected quote changes from the actual quote changes for the nontrading exchanges over a five-second evaluation period. For example, if four exchanges (including the trading one) out of six quote the NBBO price, then the average is taken over the unexpected price responses of the three nontrading exchanges. Third, following equation (10), I compute the inventory price impact as the difference between the price responses for the trading and nontrading exchanges. Equations (9) and (10) also explicitly account for trade direction. The previous steps are computed in dollar terms. I must normalize the price impacts by the option quote midpoint of a given trade to make them comparable across options and stocks. Finally, as required by equations (9) and (10), I compute the average over all trades for the individual price impacts to compute the final estimates of the asymmetric information and inventory risk components.

[Table I here]

Summary statistics for the final sample are reported in Panel A of Table I. The asymmetric information and inventory risk impacts are 0.22% and 0.41%, respectively. Other sample statistics over the individual price impacts are also computed. The standard

deviation is larger for the inventory impact (1.6% versus 2.6%) because the information impact averages price responses from multiple exchanges for each trade. The median information impact is negative because, due to large tick size, quoted prices do not change after most trades while expected quote changes are mostly positive (quotes are expected to move in the trade direction). This difference between the mean and median caused by price discreteness is further explained in the Internet Appendix.

On average, more than four exchanges out of six are quoting the best price at the time of a trade. The distribution for trade size is close to exponential with a mean of 29 contracts (or about 3,700 dollars) and a median of eight contracts (800 dollars). Consistent with the literature on equity options, call trades outnumber put trades (66%) and there are more seller-initiated trades (58%), but mostly in call options. Option and stock prices vary substantially, justifying normalization of price impacts.

IV. Empirical Results

The method that I introduce yields five main empirical results. First, contrary to previous literature (Vijh (1990)), I find that option trades have a large price impact. The asymmetric information impact is 0.22% and the inventory risk impact is 0.41% for an average trade.¹¹ Thus, both asymmetric information and inventory risk contribute to price formation. Table II shows that the asymmetric information component is positive and significant for every stock in the sample. Except for two special cases, America Online (0.02% in a small sample) and QQQ Nasdaq ETF (0.04%), the remaining values are large and lie between 0.13% (Nextel) and 0.5% (Ford). The price impacts remain positive for most trade subsamples based on option and trade characteristics.

[Table II here]

Second, the inventory risk component is large for each stock in the sample – Dow Jones SPDR ETF has the smallest impact of 0.19% while Bristol Myers has the highest impact of 1.0%. Also, inventory risk has a larger price impact than asymmetric information for every stock in my sample. Thus, option market-makers in my sample are more concerned about managing inventory risk than about trading against informed investors. Further, the inventory impact is larger for any trade size and moneyness as shown in Figures 2 and 3. The difference is particularly large for small trades and out-of-the-money options. These results suggest that inventory risk is of first-order importance and thus deserves more research attention.

[Figure 2 here]

Third, both price impacts are monotonically increasing in trade size. The nonparametric estimates in Figure 2 show that the inventory impact is linear with some concavity while the information impact is clearly concave. This result is formally confirmed in multivariate analysis that accounts for both linear and square root terms for trade size. The results, reported in Tables III and IA.IX, support the square root as an appropriate functional form here. The asymmetric information impact for large orders (70 lots) is about five times larger (0.1% versus 0.5%) than for small orders (3 lots), while the increase for the inventory impact is only twofold from 0.3% to 0.65%. The price impacts increase less steeply for large trades (more than 100 lots), resulting in a pronounced concavity for trades of this size. However, while many theories predict that the asymmetric information impact should be increasing in trade size, this result needs to be reconciled with Anand and Chakravarty (2007), who find evidence of stealth trading in the options market. In particular, they find that the Hasbrouck's information share is

twice as high for mid-sized trades as for large trades (more than 99 contracts). There is little need for stealth trading in the options market today, however, since order flow is sparse and is spread across multiple contracts, making it hard for informed traders to hide by splitting orders into mid-sized pieces. A possible explanation is that Anand and Chakravarty (2007) use data from 1999, when options were traded manually on the floor with no linkage between exchanges.

[Table III here]

Fourth, stock prices respond instantly to option trades. This result confirms that option trades do indeed contain private information about the underlying price and therefore complements existing literature, which is based mostly on daily return data. The underlying price moves by 0.02 basis points (or 0.03 cents in dollar terms) in the direction of the option trade (after adjusting for option type; for example, the stock price decreases after one buys a put). Although the effect is not large economically, it is highly statistically significant and is positive for all but one stock (Nasdaq ETF QQQ). The last column in Table III reports how the stock price impact of option trades depends on option characteristics. The single most significant variable is trade size. Large trades have three times more impact than small trades. Surprisingly, trades of different moneyness have similar price impact, while short-term options have slightly higher price impact. Overall, although option trades contain significant information about the underlying price, most of their information is option-specific.

Table III reports how the price impact components depend on option and trade characteristics.¹² Out-of-the-money options offer the highest leverage (exposure for a dollar invested) and thus are particularly attractive for informed investors. Consistent

with this argument, the information price impact is decreasing and convex in absolute delta. Figure 3.D shows that the impact decreases from 0.4% for out-of-the-money options to 0.15% for in-of-the-money options. Next, private information is often short-lived and is related to near-term events, and thus short-term options are better suited for informed investors in addition to providing higher leverage. Indeed, the price impact decreases by 0.12% if time-to-expiration decreases from 80 days to 20 days. Buyer-initiated trades have a higher price impact than sell trades, because these trades provide an opportunity to bet not only on future volatility but also on the underlying direction. These results are broadly consistent with Pan and Poteshman (2006), except that I do not find a significant difference between call and put options, perhaps because my sample consists of large stocks that are easy to sell short. As for inventory risk, its price impact also decreases with moneyness, from 0.8% for out-of-the-money to 0.2% for in-the-money options. Put-call parity implies that the paired in-the-money and out-of-the-money options should have similar risk profile, but because the price impact for out-of-the-money options is normalized by a smaller option price, this mechanically leads to larger price impact for them. Buy trades have a much larger inventory impact (0.15%) because a market-maker selling an option faces large downside risk. The inventory risk impact is higher for calls than for puts because a seller of a call faces the risk of unlimited losses. Overall, the inventory and information impacts have similar conditional properties because the options with the highest leverage also have the highest inventory risk per dollar invested.

[Figure 3 here]

Several additional tests are reported in Figure 3 and in the Internet Appendix. Earnings announcement days are characterized by high intraday volatility and information asymmetry. Indeed, both price impacts are higher on announcement days. The time of day has similar implications as volatility, which is high at the beginning and end of trading. Consistent with this hypothesis, both price impacts are higher during the first and last hours of trading. I also find that the information impact increased significantly during the sample period, while the inventory impact remained largely unchanged. To check the robustness of the nonparametric estimates of price impacts in Figure 2, I examine several subsamples in Figure 3 and confirm that inventory risk continues to have a larger price impact than asymmetric information. The first panel shows the price impacts for a sample in which the absolute options delta is in the narrow range between 0.4 and 0.6, and time to expiration is less than 50 days, to confirm the results in a homogeneous set of options. Next, I consider a sample in which the number of quoting exchanges is exactly three instead of at least two to confirm that the results hold if the number of exchanges is fixed. Finally, I increase the evaluation period is increased from five to 10 seconds in the last subsample.

It is worth repeating that because my sample consists mostly of stocks with liquid options, the results are hard to extrapolate to stocks with illiquid options. Easing this concern, both the information and inventory price impacts are larger for stocks that had the lowest average option trading volume during the periods in which they were in the sample.¹³ In addition, my results offer little help in answering why option bid-ask spreads are so large. This is one of the biggest puzzles in the options literature – existing theories of the option spread fail to explain its magnitude and shape (Muravyev and Pearson

(2014)). For example, the effective bid-ask spread is almost 7% in my sample of trades. I find that although the inventory risk and asymmetric information components are large, they explain only 18% of the spread size (and thus the remaining 82% is attributed to fixed costs). These estimates are consistent with the decompositions for the stock spread. For example, Huang and Stoll (1997) attribute 88.6% of the stock spread to the fixed costs component.

V. Inventory Risk and Daily Option Returns

The microstructure approach provides extensive evidence on the role of inventory risk in option price formation. However, this approach is not suitable for answering a number of questions. For instance, how large is the effect of inventory risk at the daily and longer horizons, and while the effect of inventory risk on prices is by definition temporary, how long does it last? Also, the tick-level options data set is hard to process because of large size, which constrains microstructure approaches to a small number of stocks (39 in my case). In this section, I take an alternative approach to measure inventory risk at the daily level, which addresses these questions and complements my intraday results above.

Specifically, having established the importance of inventory risk at the intraday level, in this section I provide support for this conclusion at the daily level with two pieces of evidence. First, using the full panel of option daily returns, IV estimation shows that the inventory-related part of option order imbalances has a five times larger impact on option prices than previously thought. Second, using a horse-race between option return predictors, I show that order imbalances have more predictive power than 50 other plausible predictors of future option returns.

Prior literature provides evidence on the correlation between order imbalances and implied volatility. The current paper complements this evidence by identifying the inventory-related part of order imbalances and quantifying its causal effect on option returns. More importantly, the paper shows that inventory risk is not a secondary factor as previously thought but is a first-order determinant of expected option returns. Bollen and Whaley (2004) are the first to emphasize the importance of order flow for option prices. They show that daily changes in the level and slope of implied volatility are positively correlated with the same-day option order imbalance. Similarly, Pedersen, and Poteshman (2009) show that the market-maker net positions are correlated with the spread between the observed implied volatility and the implied volatility estimated from a jump-diffusion model. Although these papers establish the importance of order flow, its magnitude is smaller than for other determinants of the option risk premium. Also, several issues complicate interpretation of their results.¹⁴ First, order flow is often informed, and thus prices can change in response to information rather than inventory risk. Indeed, several papers, most notably Pan and Poteshman (2006), provide evidence of informed trading in the options market. The other important concern is endogeneity. For example, both returns and order imbalances on a given day are determined by common factors such as news. Are investors buying because prices are rising or vice versa? This paper addresses these concerns by extracting the inventory-related component of order flow and using IV estimation to account for endogeneity. After addressing these endogeneity concerns, the effect of inventory risk on option prices is five times larger than previously thought.

A. *Computation of Option Returns and Order Imbalances*

The two key variables for the daily analysis are option returns and order imbalances. Returns of delta-neutral option portfolios assess a cumulative risk premium embedded in option prices without relying on a specific option pricing model. Delta-neutral portfolios are immune to small changes in the underlying and thus to stock risk premiums. Order imbalance is a direct measure of changes in option market-maker positions and thus their inventory risk. I divide options into four time-to-expiration groups: expiring options (with less than 13 calendar days to expiration), short-term options (with 30 days to expiration on average), mid-term options (with no more than 150 days left), and finally long-term options. For each day, stock, and expiration group, I pick the call-put pair closest to the at-the-money point for the shortest expiration. For the selected call-put pair, I compute option returns from the end of day $t-1$ until the end of the next trading day t . Specifically, a straddle portfolio is created by buying one call option and as many puts $(\Delta(C_t)/|\Delta(P_t)| \text{ puts})$ as required to make it delta-neutral and then held for one day. Following the literature, I compute returns using option quote midpoints. Equation (11) summarizes the return definitions as follows:

$$\text{Straddle: } W_t = C_t + \frac{\Delta(C_t)}{-\Delta(P_t)} P_t; \text{ Return} = \frac{W_t(t+1) - W_t(t)}{W_t(t)}. \quad (11)$$

For robustness, I also use delta-neutral returns on individual calls. Unless otherwise stated, I refer to the delta-neutral straddle returns for short-term options simply as “option returns”. Table IV shows that average option returns are close to zero in my sample. However, as expected, the return distribution is skewed and the median for daily returns is -1.3%. If the underlying price does not change, the straddle’s value decreases.

[Table IV here]

I compute option order imbalances accommodated by market-makers at the stock level and then also aggregate them into the market-wide imbalance. Following Chordia and Subrahmanyam (2004), I define the order imbalance as the difference between the number of option buy and sell transactions by nonmarket-makers divided by the total number of option trades on a given day. For example, market-makers are net sellers of options if the order imbalance is positive. Importantly, the data identify which trade side is taken by option market-makers. The order imbalance is calculated as follows:

$$Ord\ Imb_{A,t} = \frac{\sum_i (\#BuyTrades_{A,t,i} - \#SellTrades_{A,t,i})}{\sum_i \#Trades_{A,t,i}}. \quad (12)$$

This simple order imbalance can be adjusted to distinguish between low and high volume days by normalizing it by the average number of trades in the previous 30 days instead of the number of trades on day t . A measure of market-wide order imbalance indicates whether option investors are buying or selling at the aggregate level. I calculate it as the average of individual order imbalances weighted by the average option volume over the previous 250 trading days:

$$MWOrdImb_t = \sum_j \left(Ord\ Imb_{j,t} * \frac{1}{250} \sum_{i=0}^{250} OptVolume_{j,t-i} \right). \quad (13)$$

Table IV shows that this market-wide order imbalance is -5% in options on average. Thus, option investors are net sellers of equity options, especially call options.

After standard filters are applied,¹⁵ the final sample in most regressions exceeds one million stock-day observations, with more than 1000 eligible stocks on a typical day. In my analysis I control for most of the variables known to predict option returns. The Internet Appendix describes more than 50 control variables. Table IV reports summary statistics for some of the variables. Interestingly, the correlation matrix shows

that despite option volume, skew, and the bid-ask spread being often used as proxies for demand pressure, they have a negligible correlation with a direct measure of order imbalance.

B. Instrumental Variables Approach

An IV approach resolves the endogeneity between concurrent option returns and order flow by establishing that past order imbalances are good instruments of future imbalances. Past order imbalances predict future changes in market-maker inventory, which in turn cause changes in option prices. This idea is implied by Chordia and Subrahmanyam (2004) in their model.

The first stage of the 2SLS regression establishes that option order imbalances are highly persistent. In particular, future order imbalance is predicted separately by three nested sets of instruments in equation (14) below. The first set contains only expiration day dummies, with a separate dummy variable for each of five days centered on the post-expiration Monday. The second instrument set expands the first set by adding two lags of market-wide order imbalance to the expiration dummies. Market-wide imbalances are not affected by informed trading as private information about the entire market is hard to obtain. Finally, the third set of instruments adds two lags of individual order imbalance to the second set. With this instrument set, both time-series and cross-section dimensions can be explored.

Expiration dummies are particularly good instruments. Investors substitute expiring option positions with similar nonexpiring ones in the three-day window around the expiration day (every third Friday of a month). Because investors are short call and put equity options on average, the rollover creates unprecedentedly large selling pressure

in the nonexpiring options. Option expirations create exogenous variation in order imbalance, and thus exogenous variation in market-maker inventories as investors open new positions to replace positions in expiring options. Volatility and returns of the underlying stocks change little around expiration. Thus, fundamentals and informed trading are not responsible for the order imbalance.

Lagged variables are a popular instrument choice in economics, and they do indeed solve the econometric problem here. However, it is not a priori clear whether all predicted order imbalances should be attributed to the inventory risk channel. For example, individual order imbalances may be informed about future volatility and, through it, about future option returns. The private information contained in order imbalances may take time to be reflected in the prices. In the Internet Appendix I address this concern by showing that the predicted order imbalances are driven primarily by inventory risk rather than informed trading.

After settling on the instruments, I apply a standard 2SLS approach to study the effect of order imbalance on option returns, which is α_1 in equations (14) and (15). Specifically, first stage:

$$\begin{aligned} \text{AdjOrdImb}_{t,i} &= \beta_0 + \beta'_1 \text{IVSet}(I)_{t-1,i} + \beta' \text{Controls}_{t-1,i} + \varepsilon_{t,i}, \\ \text{where } I &\in \{1, 2, 3\} \text{ and instrument sets :} \\ \text{IVSet}(1) &= \{n_{-2}, n_{-1}, n_0, n_1, n_2\}; \\ \text{IVSet}(2)_{t-1} &= \{\text{IVSet}(1) ; \text{MWOrdImb}_{t-1}, \text{MWOrdImb}_{t-2}\} \\ \text{IVSet}(3)_{t-1,i} &= \{\text{IVSet}(2)_{t-1}; \text{OrdImb}_{t-1,i}, \text{OrdImb}_{t-2,i}\} \end{aligned} \quad (14)$$

second stage:

$$\text{OptRet}_{t,i} = \alpha_0 + \alpha_1 \text{AdjOrdImb}_{t,i} + \beta' \text{Controls}_{t-1,i} + \varepsilon_{t,i}. \quad (15)$$

All the coefficients in the first stage are highly significant, as reported in Panel A of Table V, and hence the instruments are strong. Order imbalance is extremely negative

around option expiration because investors are rolling over their positions to nonexpiring options. The selling pressure is particularly large on the post-expiration Monday when the abnormal order imbalance reaches -24%. Large negative order imbalances are also observed on the day before and after the post-expiration Monday (-8.7% and -11.7%, respectively). The expiration rollover is perhaps the strongest calendar seasonality in the options market. Next, order imbalances are very persistent in the options market, as past market-wide and individual imbalances have large positive coefficients. For example, a market-wide imbalance of -10% today predicts a -6.5% imbalance tomorrow. Thus, if an option market-maker observes an increase in his inventory on one day, his net position is likely to continue to increase on the next day.

[Table V here]

The second-stage results are reported in Panel B of Table V. First, the average return sensitivity to order flow across the instrument sets is 0.15, which means that inventory risk has a very large impact on option prices. For example, one-standard-deviation increase in expected order imbalance (which is 25% for the last instrument set) corresponds to a 3.7% option return. Second, the OLS coefficient of 0.025 is consistent with estimates by Bollen and Whaley (2004) after adjusting the units. However, this coefficient is six times smaller than the 0.15 produced by the IV approach. Thus, accounting for endogeneity makes a big difference. The measurement error bias can partially explain the difference in estimates.¹⁶ Third, the coefficient estimates on all the instrument sets have similar magnitudes (15.4%, 17.8%, and 11.5%, respectively), which validates the instruments. Finally, the fact that market-wide order imbalance has a significant effect on returns distinguishes the inventory channel from the “mechanical

price pressure” hypothesis. The price pressure hypothesis does not expect option prices to be moved by order imbalance in other securities.

Overall, the IV approach shows that inventory risk is a major factor for the entire panel of option daily returns.

C. Return Predictability: Inventory Risk versus Other Factors

The IV approach assesses the absolute magnitude of the inventory risk effect on option prices while this section evaluates its relative importance for predicting future option returns. The IV approach shows that past order imbalances are an excellent measure of future inventory shocks. Building on this result, I conduct a direct horse-race between the past order imbalance and more than 50 plausible option returns predictors. The order imbalance wins the contest by a large margin. Also, the horse-race ranks the relative importance of option “anomalies,” so that the literature can focus on explaining the largest of them.

I estimate a panel regression in equation (16), where a set of past variables predict the option return on the next day. As suggested by the IV approach, individual and market-wide order imbalances represent inventory risk. In the Internet Appendix I provide a complete list of more than 50 control variables that include risk-neutral volatility, implied skewness and kurtosis, past stock returns, different measures of stock volatility, measures of size such as volume and capitalization, implied volatility skew and time slope, and a weekend dummy. These variables cover most of the known option market “anomalies.” I test these variables in different combinations and select the set of the best 15 predictors that avoids collinearity.

$$OptRet_{t,i} = \alpha_0 + \alpha_1 OrdImb_{t-1,i} + \alpha_2 MWOrdImb_{t-1} + \beta' OtherPredictors_{t-1,i} + \varepsilon_{t,i}. \quad (16)$$

Based on the results in Table VI, if the combined order imbalance increases by one standard deviation, option returns are expected to be 1% higher on the next day, while other predictors have a magnitude of at most 0.4% (per standard deviation). First, I include the individual order imbalance on day $t-1$, which is highly significant with a t -statistic of 18 even in the presence of all the control variables. Next, I add the market-wide order imbalance, which has a t -statistic of “only” nine because its value is the same for all stocks on a given day, reducing its statistical power. To assess the combined economic magnitude of order imbalance measures, I aggregate them into a single factor using coefficient estimates from the regression.

[Table VI here]

Magnitudes of other predictors are mostly consistent with the literature and are smaller than for the order imbalance. Of the fifty variables, the cross-section of option returns is most affected by absolute stock returns (0.33% option return per standard deviation increase) and the option bid-ask spread (0.2%). Christoffersen et al. (2014) find a similar magnitude for the spread. The fact that my results are consistent with other theories reported in the literature indicates that my specification is reasonable. Weekend return anomaly introduced by Jones and Shemesh (2012) is -1.3% on average, which makes it perhaps the biggest calendar seasonality in option returns. Interestingly, during my sample period, the weekend effect is mostly driven by the expiration weekend and almost entirely disappears in the subsample of options with price above two dollars. Changes and levels of implied volatility can predict next-day returns (0.39%) but fail to predict returns on subsequent days and thus are likely to be driven by microstructure effects. In the Internet Appendix I discuss these and other predictors.

Overall, option order flow, a proxy for future inventory shocks, is by far the strongest predictor of future option returns.

VI. Conclusion

The paper presents three pieces of evidence showing that inventory risk has a first-order effect on option prices. First, I find that option trades have substantial price impact, and that the inventory risk component of the price impact is larger than the asymmetric-information component for each stock in my sample. This intraday result is supplemented by two results that use daily order imbalances accumulated by option market-makers. IV estimation finds that the inventory related part of option order imbalances has a five times larger impact on option prices than previously thought. Finally, a horse-race between option return predictors shows that past order imbalances have more predictive power than 50 other plausible predictors of option returns.

The daily results complement the intraday evidence in several ways. First, they show that inventory risk has a long-term effect on option prices and provide alternative ways of assessing its economic significance. For example, the inventory risk premium is larger than most known option return “anomalies.” Also, these results show that inventory risk matters for the full panel of optionable stocks and a longer sample period than is available for the intraday analysis.

The paper introduces a novel method for decomposing the price impact of trades into inventory risk and asymmetric information components and then applies it to the equity options market. The inventory risk component is inferred from the difference between the price responses of the market-maker that receives a trade and those who do not. The asymmetric information component is estimated as the average price response of

market-makers that do not participate in a given trade adjusted for expected changes in price. This method is particularly well suited for the options market. However, it can be also applied to many other markets including the bond and equity markets. Overall, this method contributes to the literature on the role and measurement of market-maker inventory risk and the literature on how to distinguish the inventory risk from the asymmetric information components of bid-ask spreads.

I apply the method above to equity options and find that inventory risk plays a dominant role in option price formation. Both information and inventory price impacts are large for option trades, but the inventory risk component is almost always larger in my sample. Thus, option trades have high information content. In particular, while option market-makers are concerned about trading against informed investors, they are even more concerned about managing inventory risk. Overall, my results suggest that inventory risk in the options market, and compensation for bearing it, are important components of option price dynamics.

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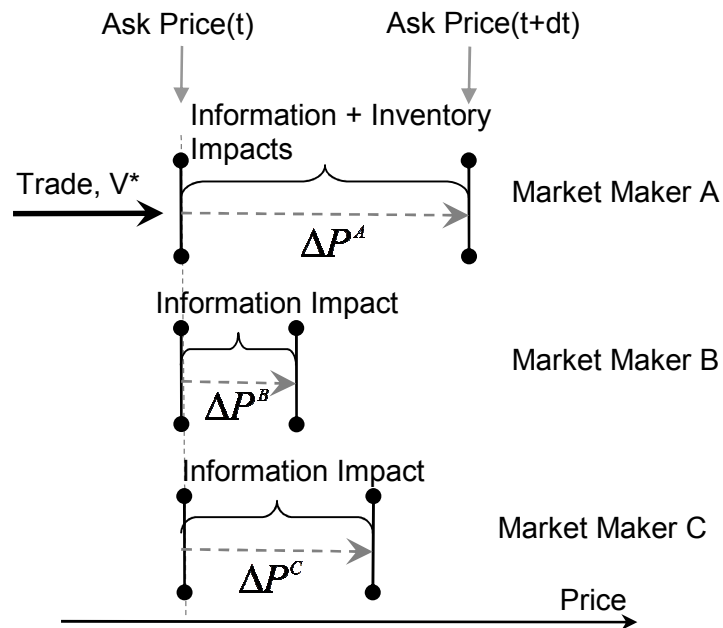


Figure 1. Stylized example of how the asymmetric information and inventory risk components of price impact are computed for a buyer-initiated trade. Three market makers (A, B, and C) are quoting the same best ask price when market-maker A receives a buy trade of size V^* . Price responses are measured between time t and $t+dt$. Market maker A responds to both asymmetric information and change in inventory, while market-makers B and C only react to trade information because their inventory does not change. For simplicity, this example assumes that the expected changes in the ask prices are zero.

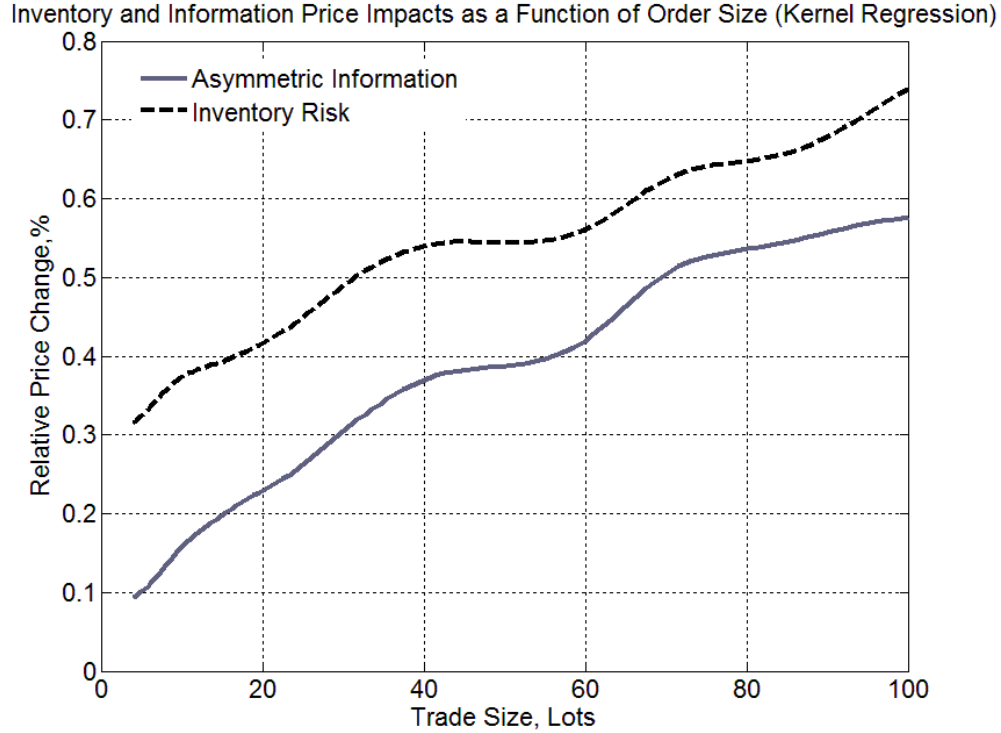


Figure 2. Nonparametric estimates of the inventory risk and asymmetric information components of price impact as a function of option trade size. The components of price impact are estimated by equations (9) and (10), are measured over a five-second evaluation period, and are normalized by the option quote midpoint. All price impacts and expected quote changes are adjusted to make the sign comparable across buys and sells. A kernel regression uses the Gaussian kernel with an adaptive width that is larger in the low-populated regions (big trade size). Each option contract is written on 100 underlying shares. Confidence intervals are not reported because they are trivial with a large sample size.

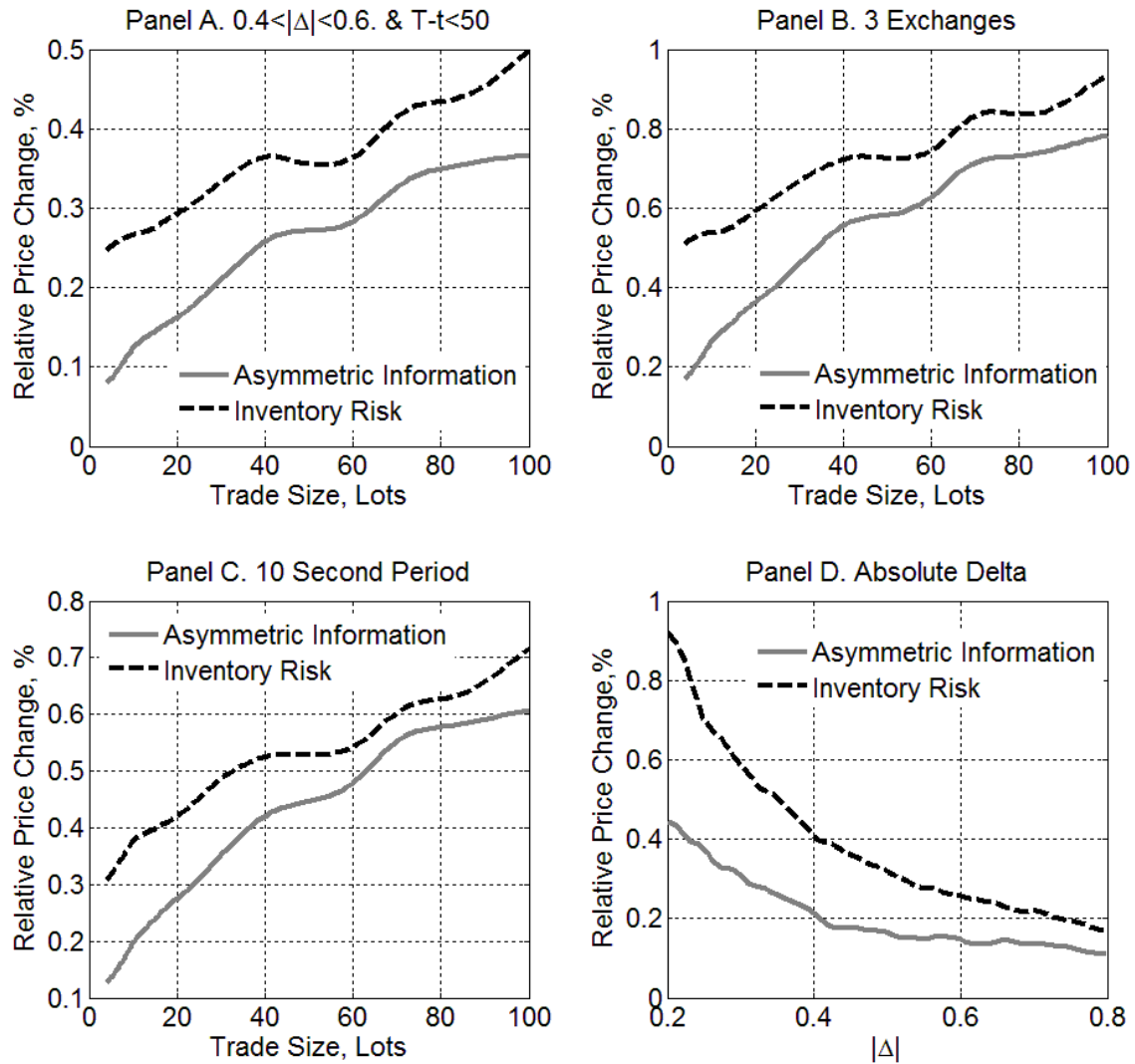


Figure 3. Robustness checks for nonparametric estimates of inventory and information price impacts as a function of trade size. The price impacts are computed according to equations (9) and (10). Panel A reports the price impacts for the subsample of trades that are at-the-money and have less than two months to expiration. Panel B shows the price impacts for trades with exactly three exchanges quoting the best price in the trade direction at the time of a trade. Panel C uses the evaluation period of 10 seconds instead of five seconds for the main sample. Panel D reports how the price impacts depend on moneyiness measured as absolute option delta.

Table I
Summary Statistics and Data Filters

Panel A reports summary statistics (average, standard deviation, and percentiles) for the final sample of option trades. The components of price impact are estimated by equations (9) and (10) over five-second evaluation period and are normalized by option price. All price impacts and expected quote changes are adjusted to make the sign comparable across buys and sells. The price impact of option trades on the underlying stock is adjusted for trade direction and is normalized by stock quote midpoint. *Number of Exchanges at NBBO* is computed in the direction of a trade (e.g., ask for buy). Each option contract is written on hundred underlying shares. Panel B reports the sample size after each of the filters applied in constructing the final sample.

Panel A. Summary Statistics

Variable	Mean	Std. Dev.	5%	50%	95%
Information Price Impact, %	0.22	1.58	-0.36	-0.03	2.01
Inventory Price Impact, %	0.41	2.56	-0.75	0.00	4.08
Stock Price Impact, bp	0.02	0.60	-0.79	0.00	0.89
Expected Quote Changes, %	0.08	0.42	-0.1	0.04	0.41
Trade Size, Contracts	28.97	154.80	1	8	100
Trade Size, 100s \$	36.79	261.90	0.6	8	131
Buy/Sell Dummy	0.42	0.49	0	0	1
Call/Put Dummy	0.66	0.47	0	1	1
Option Price, \$	1.56	1.53	0.28	1.08	4.45
Stock Price, \$	39.89	20.91	14.23	35.68	84.88
Implied Volatility	0.28	0.12	0.14	0.25	0.48
Absolute Delta, $ \Delta $	0.46	0.15	0.23	0.45	0.74
Days to Expiration	80.68	78.87	15	46	246
Number of Exchanges at NBBO	4.18	1.41	2	4	6

Panel B. Filters and Sample Size: Number of Trades after each Filter

Total	28,127,906
0.2 < $ \Delta $ < 0.8	
10 < Days-to-expiration < 400	15,298,816
9:35 < Time < 15:55	
Trade price = NBBO price, with at least 2 exchanges quoting it, Trade size < quoted size, Option price > 10 cents	7,684,944
Outliers for price impact	7,684,040

Table II
The Components of Price Impact for the 36 Stocks and Four ETFs

Asymmetric information and inventory risk price impacts as well as expected changes in option price and the price impact of option trades on the underlying are reported for each stock in the sample. All price impacts are adjusted to make the sign comparable across buys and sells. Ticker symbols indicated with * were dropped before the end of the sample period, while those indicated with ** were added after the start of the sample. All price impacts are in % except for the stock price impact, which is in basis points.

Stock Ticker	# Obs.	Information Impact	Inventory Impact	Expected Change	Stock Price Impact
AIG	116,845	0.21	0.28	0.07	0.024
AMAT	150,655	0.24	0.62	0.08	0.026
AMGN	180,918	0.20	0.31	0.10	0.028
AMR	94,975	0.36	0.52	0.06	0.061
AMZN	217,455	0.21	0.31	0.10	0.054
AOL	21,271	0.02	0.73	0.04	0.012
BMJ	82,125	0.34	1.00	0.06	0.011
BRCM	175,646	0.22	0.28	0.09	0.053
C	195,536	0.24	0.35	0.07	0.007
COF	58,791	0.21	0.35	0.07	0.043
CPN*	50,278	0.44	0.73	0.05	0.044
CSCO	320,531	0.23	0.42	0.07	0.014
DELL	185,196	0.28	0.37	0.09	0.025
DIA**	100,969	0.18	0.19	0.05	0.008
EBAY	422,516	0.17	0.21	0.10	0.034
EMC	106,407	0.31	0.74	0.06	0.016
F	65,093	0.50	0.87	0.05	0.022
GE	262,511	0.21	0.43	0.06	0.004
GM	199,507	0.31	0.44	0.07	0.027
HD	147,018	0.18	0.40	0.06	0.011
IBM	250,684	0.14	0.23	0.08	0.011
INTC	475,534	0.22	0.42	0.08	0.009
JPM	148,222	0.25	0.43	0.07	0.014
KLAC	93,175	0.22	0.31	0.12	0.039
MMM	91,807	0.20	0.28	0.07	0.020
MO	181,345	0.18	0.27	0.06	0.012
MSFT	368,470	0.26	0.56	0.07	0.011
MWD*	53,712	0.23	0.51	0.08	0.014
NXTL*	67,671	0.13	0.44	0.07	0.030
ORCL	118,254	0.39	0.62	0.07	0.025
PFE	289,798	0.28	0.51	0.07	0.005
QCOM	256,123	0.22	0.25	0.11	0.036
QLGC	80,366	0.24	0.31	0.09	0.048
QQQ*	824,710	0.04	0.31	0.10	-0.003
QQQQ**	678,154	0.26	0.55	0.13	0.006
SBC*	46,229	0.22	0.88	0.06	0.007
SMH	162,516	0.23	0.40	0.08	0.022
TYC	88,229	0.32	0.45	0.05	0.017
XLNX	54,253	0.31	0.66	0.11	0.047
XOM	200,545	0.19	0.44	0.06	0.013
Mean or sum	7,684,040	0.22	0.41	0.08	0.018

Table III**Price Impacts Conditional on Trade and Option Characteristics**

The components of price impact are estimated by equations (9) and (10). Nonlinearity in trade size is accounted for by including the square root in addition to a linear term. Nonlinearity in absolute option delta is accounted for by including variables equal to the absolute delta for particular moneyness groups (out-of-the-money or at-the-money) and zero otherwise. The absolute t -statistics reported in parentheses are based on robust standard errors clustered by date. Stock-date fixed effects are included but not reported.

	Information Impact, %	Inventory Impact, %	Stock Price Impact, bp
Absolute Delta, $ \Delta $	-0.746 (22.25)	-1.775 (42.77)	0.007 (1.52)
$ \Delta $, if $ \Delta < 0.4$	-0.110 (5.84)	-0.347 (13.04)	0.001 (0.28)
$ \Delta $, if $0.4 \leq \Delta < 0.6$	-0.097 (12.64)	-0.227 (21.93)	-0.001 (0.35)
$\sqrt{\text{Days to Expiration}}$	-0.027 (23.30)	-0.050 (40.54)	-0.001 (6.36)
Call/Put Dummy	-0.000 (0.05)	0.073 (16.37)	-0.002 (2.85)
Option Price, \$	0.041 (17.36)	0.076 (24.38)	0.003 (5.72)
Buy/Sell Dummy	0.033 (5.44)	0.159 (27.53)	0.002 (2.72)
$\sqrt{\text{Trade Size}}$	0.032 (40.92)	0.018 (21.68)	0.005 (36.69)
Trade Size, Contracts	-0.000 (16.57)	-0.000 (11.05)	-0.000 (2.84)
R^2	0.04	0.03	0.01
# Obs.	7,684,040	7,684,040	7,684,040

Table IV
Summary Statistics for Daily Data

This table reports the mean, standard deviation, and median as well as 10th and 90th percentiles for selected variables. Options are divided into four days-to-expiration groups: ultra short-term ($p=0$) with less than 13 calendar days to expiration; short-term ($p=1$) with on average 30 days; mid-term ($p=2$) with no more than 150 days; and long-term ($p=3$) with more than 150 days. Option returns (*OptRet*) are computed for a delta-neutral straddle portfolio (long) based on the call-put pair that is closest to at-the-money. The order imbalance is based on the difference between the number of buy and sell trades normalized by the total number of trades on a given day (*OrdImb*) or by the average number of trades in the previous 30 days (*AdjOrdImb*). *MWOrdImb* is the market-wide order imbalance. *OptBidAsk* is the dollar option bid-ask spread for short-term options. *OptVolume* is option volume measured in contracts. *IV30* is 30-days-to-expiration implied volatility. *Skew60* is the logarithm ratio of out-of-the-money to at-the-money implied volatilities for 60-days-to-expiration options. *RNSkewness* is risk-neutral skewness. *Diff(IV)* is the one-day change in short-term implied volatility. *IV60 - StdRet* is the difference between implied and historical volatilities. Complete variable definitions can be found in the Internet Appendix.

	# Obs (in 1,000s)	Mean	Std.Dev.	P10%	Median	P90%
OptRet($p=0$)	606	-0.010	0.505	-0.400	-0.059	0.431
OptRet, OptRet($p=1$)	1,735	-0.000	0.096	-0.071	-0.013	0.077
OptRet($p=2$)	1,819	0.000	0.051	-0.038	-0.004	0.041
OptRet($p=3$)	1,817	0.001	0.033	-0.025	-0.001	0.028
OrdImb	1,813	-0.054	0.271	-0.400	-0.045	0.292
AdjOrdImb	1,813	-0.047	0.478	-0.442	-0.033	0.326
MWOrdImb	1,771	-0.058	0.045	-0.115	-0.057	-0.003
OptBidAsk, \$	1,805	0.196	0.225	0.050	0.150	0.350
OptVolumeUSD, \$	1,841	22340	293928	84	1421	26348
OptVolume, #	1,841	7295	48603	64	845	11917
IV30	1,841	0.444	0.221	0.218	0.400	0.723
diff(IV)	1,807	0.000	0.042	-0.034	-0.001	0.037
Skew60	1,841	0.109	0.108	0.018	0.098	0.208
RNSkewness	1,841	-0.600	0.432	-1.090	-0.559	-0.154
IV60 - IV360	1,841	0.025	0.070	-0.027	0.011	0.091
IV60-StdRet	1,841	-0.020	0.161	-0.165	0.004	0.100
StkRet	1,841	0.000	0.032	-0.032	0.000	0.032
AbsStkRet	1,841	0.021	0.024	0.002	0.014	0.047
StkPrice, \$	1,841	40.09	65.97	10.05	29.51	70.96
logME	1,825	21.99	1.66	20.00	21.84	24.14

Table V
The Instrumental Variables Approach

This table reports both stages of two-stage least squares from equations (14) and (15). I use three nested sets of instruments. The first contains only expiration dummies. The second adds two lags of the market-wide order imbalance by nonmarket-makers to the first set. The third adds two lags of the individual order imbalance. $n0$ is a dummy that corresponds to the first day after expiration, usually the fourth Monday of a month (n_1 precedes $n0$). The order imbalance is based on the difference between the number of buy and sell trades normalized by the total number of trades on a given day ($OrdImb$) or by the average number of trades in the previous 30 days ($AdjOrdImb$). $MWOrdImb$ measures the market-wide order imbalance. All regressions include a battery of control variables. Absolute t -statistics reported in parentheses are based on robust standard errors clustered by date.

Panel A. First Stage			
Instrument Set:	AdjOrdImb _t		
	1	2	3
n_2	-0.018 (3.94)	-0.008 (1.92)	-0.007 (1.86)
n_1 , Friday	-0.073 (10.94)	-0.058 (9.59)	-0.057 (9.21)
$n0$, Monday	-0.234 (17.07)	-0.189 (15.19)	-0.189 (14.85)
$n1$	-0.105 (9.50)	-0.023 (2.73)	-0.018 (2.15)
$n2$	-0.056 (11.78)	0.002 (0.29)	0.005 (0.76)
$MWOrdImb_{t-1}$		0.670 (6.67)	0.569 (5.69)
$MWOrdImb_{t-2}$		0.220 (4.00)	0.117 (2.09)
$OrdImb_{t-1}$			0.173 (37.03)
$OrdImb_{t-2}$			0.086 (36.09)
<i>Other Controls</i>	+	+	+
R^2	0.02	0.03	0.04
N (in 1000s)	1,133	1,128	1,093

Panel B. Second Stage and OLS				
Instrument Set:	OptRet _t			OLS
	1	2	3	
AdjOrdImb _t	0.159 (8.54)	0.178 (10.36)	0.115 (12.05)	0.025 (31.56)
<i>Other Controls</i>	+	+	+	+
N (in 1000s)	1,118	1,112	1,094	1,080

Table VI
Horse-Race between Option Return Predictors

Option returns on day t are predicted by order imbalances and other variables from day $t-1$. Column (1) reports standard deviations to facilitate computing economic magnitudes. Columns (2) and (3) gradually add order imbalances from day $t-1$. If two measures of order imbalance are combined into one factor weighted by estimated coefficients ($0.008*OrdImb_{t-1}+0.198*MWOrdImb_{t-1}$), this factor has a standard deviation of 1%. Thus, a one-standard deviation change in the combined order imbalance changes next-day expected returns by 1%. I report the most significant control variables. *Weekend* dummy is one if t is Monday. $IV(p=1)$ is implied volatility for short-term options. $Diff(IV)_{t-1}$ is the one-day change in short-term implied volatility. $AbsStkRet_{t-1}$ is absolute stock returns. $COStdRet - StdRet$ is open-close volatility relative to close-to-close volatility. Absolute t -statistics reported in parentheses are based on robust standard errors clustered by date.

	Std. Dev.	OptRet _t	OptRet _t
	(1)	(2)	(3)
OrdImb _{t-1}	0.27	0.012 (18.83)	0.008 (16.12)
MWOrdImb _{t-1}	0.04		0.198 (9.39)
Weekend	0.40	-0.013 (4.20)	-0.013 (4.47)
OptRet _{t-1}	0.10	-0.027 (4.23)	-0.040 (6.57)
diff(IV) _{t-1}	0.04	-0.079 (5.18)	-0.088 (5.87)
AbsStkRet _{t-1}	0.02	0.142 (4.92)	0.165 (5.94)
IV(p=1) _{t-1}	0.23	-0.014 (2.73)	-0.016 (3.04)
COStdRet - StdRet	0.03	-0.020 (2.59)	-0.025 (3.33)
<i>Other Controls</i>		+	+
R^2		0.02	0.02
N (in 1000s)		1,161	1,161

Notes

¹ See Bollen and Whaley (2004) and Garleanu, Pedersen, and Poteshman (2009).

² Pan and Poteshman (2006) attribute all order imbalance to informed trading, Bollen and Whaley (2004) and Garleanu, Pedersen, and Poteshman (2009) to inventory risk, and Chen, Joslin, and Ni (2014) to fundamentals.

³ Popular methods to measure asymmetric information include Glosten and Harris (1988), Hasbrouck (1991), George, Kaul, and Nimalendran (1991), Lin, Sanger, and Booth (1995), Madhavan, Richardson, and Roomans (1997), and Huang and Stoll (1997). Madhavan and Smidt (1993), Hasbrouck and Sofianos (1993), and Naik and Yadav (2003) use actual inventory data to study market-maker behavior in the equity market.

⁴ The Internet Appendix is available in the online version of the article on the Journal of Finance website.

⁵ My assumption of fixed trade size is consistent with much of the previous literature, including Madhavan, Richardson, and Roomans (1997), Glosten and Milgrom (1985), and Huang and Stoll (1997), among many others.

⁶ For example, Madhavan and Smidt (1993) derive this condition from micro foundations in a multi-period model with a monopolistic market-maker facing both inventory risk and asymmetric information.

⁷ To better understand the multi-period price dynamics, consider a sequence of buyer-initiated trades. Imagine that three market makers quote the same best ask price and three buy trades arrive one after another. After each trade, the trading market maker increases the ask price more than the nontrading peers. After the first trade, the number of exchanges at the best asks decreases from three to two, and this number further shrinks to one after the second trade. But after the third trade, each market-maker has received one trade and all three exchanges quote the best ask price again (the price will be higher of course because of inventory and information price impacts). This example illustrates how market-makers maintain comparable inventory positions without direct interdealer trading.

⁸ This method can be potentially applied even in a market where market-makers have private information about the trades they receive. Other market-makers can learn this information by observing the price response of the trading market-maker and inferring the asymmetric information component from it.

⁹ However, if the number of market-makers is small, the interaction shows up through the second-order effect on expected order flow. The price response of the trading exchange changes the state of the limit order book. The nontrading market-makers will respond to this change because conditional on them getting the next trade, the expected time it takes for them to trade out of the extra inventory depends on the state of the book. The direction of this effect is unclear though. However, if there are many market-makers, the effect is small because actions of a single market-maker change the limit order book little.

¹⁰ Some stocks were dropped before the end of the sample period because of mergers (America Online, Calpine, Nextel, SBC) or ticker changes (Morgan Stanley, Nasdaq 100 ETF QQQ). The sample size for these stocks is relatively small, which may lead to nonrepresentative results. Two ETFs were added to the sample after the beginning. Initially, the method cannot be applied to the Dow Jones ETF (DIA) because it was traded only at a single exchange, but it was added to the sample after other exchanges started trading its options. Nasdaq 100 ETF (QQQQ) changed ticker and exchange listing; I report its results separately before and after the ticker change.

¹¹ Although not directly comparable, these numbers are also larger than typical price impacts in the stock market. Also, these price impacts are large not only in absolute value but also compared to typical abnormal option returns found for option “anomalies,” which are typically less than 1% per day.

¹² Large sample size is necessary to estimate the price impacts because the distribution for quote changes is discrete and highly skewed, with the majority of trades having zero impact. This discreteness is caused by a large tick size in the options market. R^2 is low for the same reason. All regressions are estimated with stock-day (or stock) fixed effects, while kernel regressions used for the figures are based on the pooled sample.

¹³ The five stocks with the lowest average daily option trading volume when they were in the sample are Capital One Financial, Calpine, Ford, SBC Communications, and Xilinx, with ticker symbols COF, CPN, F, SBC, and XLNX, respectively.

¹⁴ Both of these papers are based predominantly on pre-2000 data. The equity options market experienced a major change from floor-based to electronic trading after ISE entered the scene in May 2000.

¹⁵ Stock-days with stock prices below five dollars or with positive option volume on less than 80 days per year are excluded. Returns are computed only for options that satisfy three criteria: (1) both option bid prices should be larger than 10 cents, (2) OptionMetrics deltas should be well defined, and (3) the option bid-ask spread should be less than 50% of the price and the quote midpoint should be larger than 40 cents.

¹⁶ I show that the market-wide order imbalance is of central importance for option returns. If option returns are determined by market-wide order imbalance, but it is substituted in the return regression by a noisy proxy such as individual order imbalance, then the coefficient will be biased. Indeed, individual imbalance is an unbiased estimate of the market-wide imbalance but has five times larger standard deviation, which is large enough to cause a significant measurement error bias in a regression.